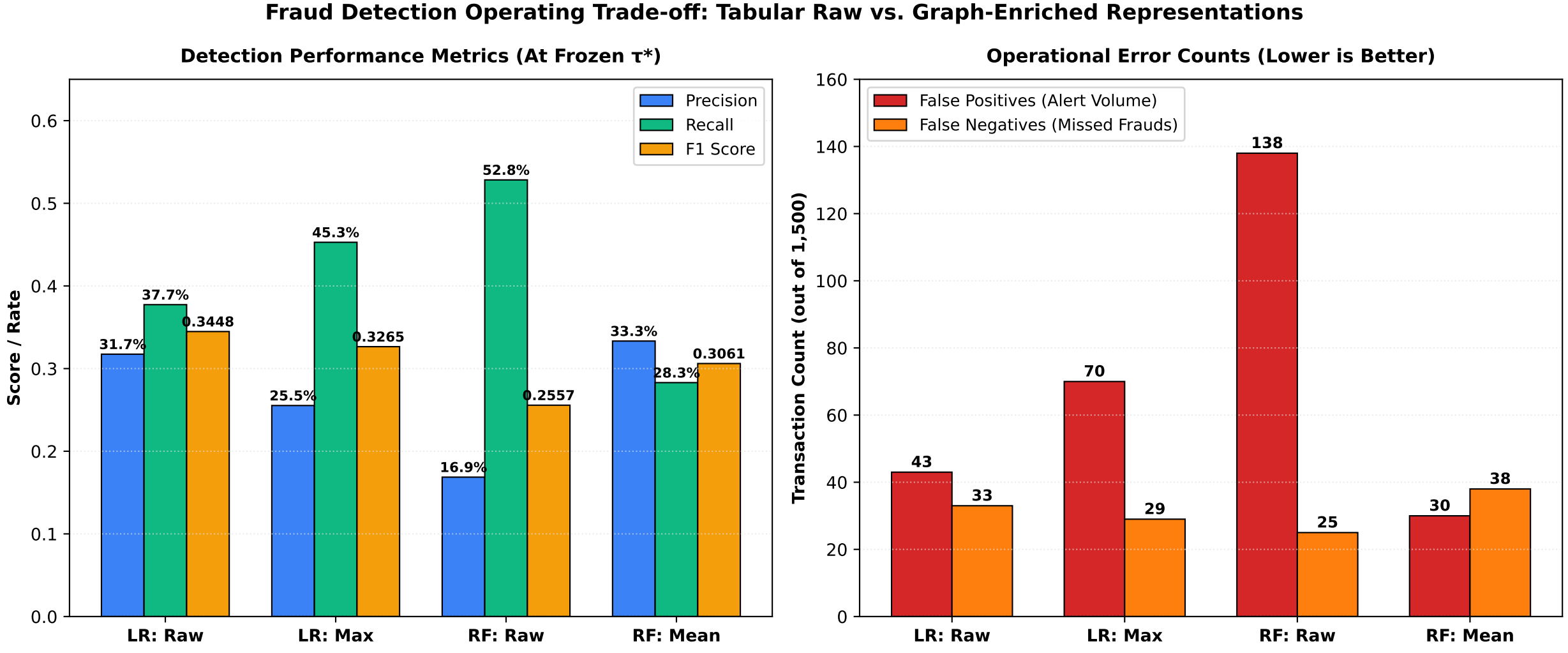
Fraud Detection Operating Trade-off: Tabular Raw vs. Graph-Enriched Representations
Detection Performance Metrics (At Frozen τ*)
Score / Rate
Precision
Recall
F1 Score
31.7%
37.7%
0.3448
25.5%
45.3%
0.3265
16.9%
52.8%
0.2557
33.3%
28.3%
0.3061
LR: Raw
LR: Max
RF: Raw
RF: Mean
Operational Error Counts (Lower is Better)
Transaction Count (out of 1,500)
False Positives (Alert Volume)
False Negatives (Missed Frauds)
43
33
70
29
138
25
30
38
LR: Raw
LR: Max
RF: Raw
RF: Mean